

Lab Update: OpenHands Agent Taxonomy

Experiment

Implementation & QA Layers

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Overview

Hi sir,

I wanted to share the results of my practical experiment testing the AI agent taxonomy using the OpenHands framework. I successfully got the local Docker sandbox running and tested the agent's ability to seamlessly shift from a Coder to a Debugger.

Here is a step-by-step breakdown of how I set up the environment and executed the test:

1 Environment Setup & Troubleshooting

Getting the local sandbox running correctly took some configuration. I initially tried running the Docker container straight from the Windows command prompt, but the OpenHands internal Linux system couldn't parse the Windows file paths (it crashed with a "too many colons" error when trying to mount my workspace).

To fix this, I completely shifted the environment over to Ubuntu via WSL. I ran into a few other roadblocks, including Linux file permission lockouts and API rate limits on the free Google tier, but I resolved them by using `chown` to reclaim my workspace directory and breaking my prompts into smaller steps so the API wouldn't time out.

2 The Experiment Design (The Trap)

My goal was to test the Quality Assurance (QA) layer. Instead of just asking it to write basic code, I set a deliberate trap to see if it could catch and fix its own runtime errors without human help. I gave the agent these instructions:

Create a file named trap.py. Write Python code that sets $A=10$, $B=0$, and prints A divided by B . Then, run the script in the terminal, read the error message, and rewrite the code to handle that error safely.

3 Execution (Implementation Layer)

The agent successfully processed the prompt, created `trap.py` in my local workspace, and wrote the hardcoded script. When it executed the file, it immediately hit a `ZeroDivisionError` in the terminal, exactly as planned.

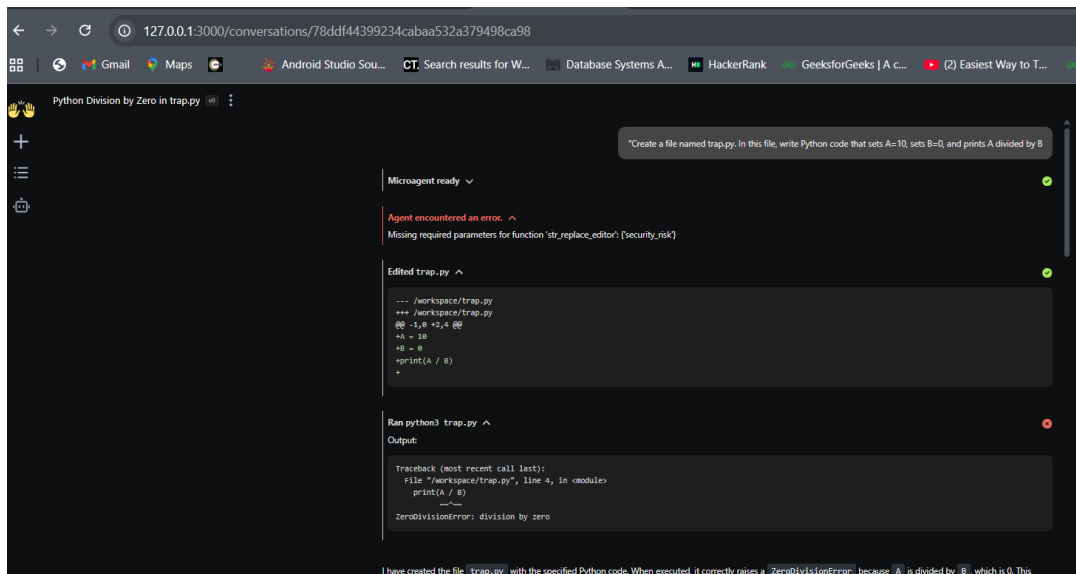


Figure 1: The agent writing the code and hitting the `ZeroDivisionError`.

4 The Fix (QA / Debugging Layer)

This is where the agent proved the taxonomy framework works in practice. It didn't just crash or pause to ask me what to do next. It autonomously read its own terminal output, recognized the `ZeroDivisionError`, and rewrote the script using a `try...except` block to handle the math safely. It essentially acted as its own tester and debugger.

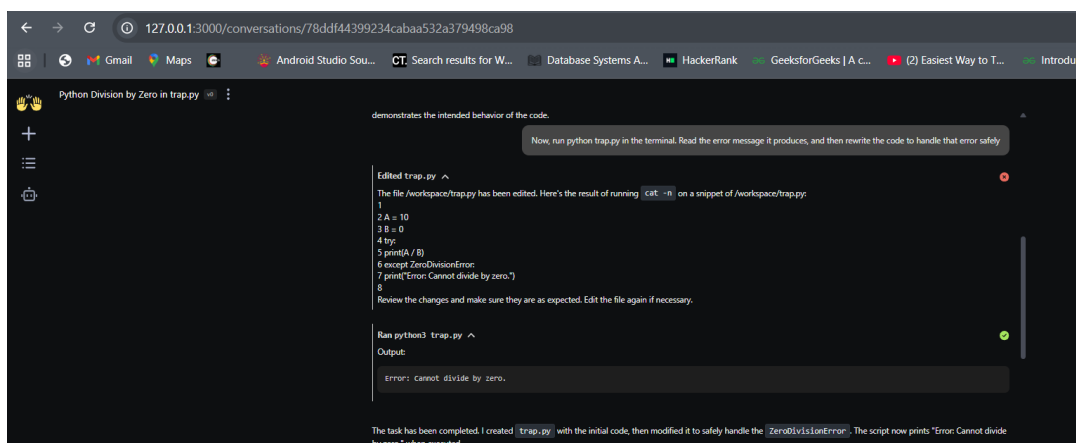


Figure 2: The agent autonomously debugging the code using a `try-except` block.

Conclusion

This test confirms that the OpenHands framework can successfully handle the QA feedback loop completely internally. I'm looking forward for your answer thanks.

Best regards,
Shahab Ali